ECHO CHAMBER BREAKER

Product Definition & Developer Knowledge Base

What the product is • Problem it solves • Deep developer context

Document Type	Product Definition + In-Depth Developer Knowledge
Audience	Solo developer / small team building the product
Purpose	Single source of truth for product intent, user value, and domain knowledge
Related Docs	Technical Architecture PDF + Pitch Deck

1. What the Product Actually Is

Echo Chamber Breaker is a self-hosted Model Context Protocol (MCP) server that turns Alexa+ into an intellectual honesty coach. When a user expresses a strong opinion, belief, or moral judgment, the system detects the strength of that stance, generates a high-quality steelmanned opposing argument, tracks whether the user actually engages with the counter-view, and maintains a private Intellectual Honesty Score over time.

Core Identity

- It is not a debate bot that tries to win arguments.
- It is not a fact-checker that simply labels statements true/false.
- It is a disciplined sparring partner whose only job is to prevent the user from living inside an unchallenged opinion bubble — including the bubble created by sycophantic AI assistants.
- It lives as tools that Alexa+ (or any MCP client) can discover and call over Streamable HTTP.

One-Sentence Definition

Echo Chamber Breaker is the anti-sycophant layer for Alexa+: it forces high-quality intellectual friction and measures whether you actually use it.

2. The Problem It Solves

The Human Problem

People (and the AIs they talk to) naturally drift toward confirmation. Strong opinions get reinforced, opposing evidence is dismissed, and over time the user becomes more certain and less accurate. Classic echo chambers were social. The new version is conversational AI that agrees too readily.

The AI-Specific Problem

Most modern assistants are optimized for helpfulness and agreeableness. When a user states a confident opinion, the default response is soft agreement or mild hedging. This feels good in the moment and is terrible for long-term reasoning quality. There is almost no mainstream voice interface that systematically challenges the user with the strongest possible opposing case and then tracks whether the user engaged with it.

Why Voice + MCP is the Right Surface

- Voice is natural for argument and reflection.
- Alexa+ already acts as an MCP client — tools can be discovered and invoked without a custom skill API.
- A web simulator can demonstrate the full experience without partner-preview access.
- The product creates a measurable signal (Intellectual Honesty Score) that is rare and interesting.

3. Who It Is For & What Value It Creates

Primary User

Curious, opinionated people who value intellectual growth over comfort: students, professionals, writers, founders, and anyone tired of both social media echo chambers and overly agreeable AI.

Jobs To Be Done

When I…	I want to…	So I can…
Express a strong opinion	Be challenged with the strongest opposing case	Test whether my view survives contact with reality
Feel very certain about something	See my blind spots named clearly	Reduce overconfidence
Finish a heated discussion	Get scored on how honestly I engaged	Build a measurable habit of intellectual honesty
Want proactive friction	Receive a challenge drawn from my history	Keep my thinking sharp without having to initiate every time

Core Value Proposition

Most AI makes you feel smarter in the moment. Echo Chamber Breaker makes you actually harder to fool — including by yourself.

4. Product Behavior a Developer Must Internalize

The Ideal Interaction Loop

1. User states a strong opinion (voice or text)
2. System runs analyze_stance → confidence + emotion + predicted blind spots
3. System runs steelman → strongest charitable opposing argument
4. User is invited to engage (or dismiss)
5. System records engagement quality via record_engagement
6. Honesty Score updates (append-only ledger)
7. Over time: honesty_report, challenge_me, multi-turn debate sessions

Non-Negotiable Product Rules

- Never strawman. Always steelman (strongest reasonable opposing case).
- Never claim to know absolute truth. The goal is friction and engagement, not victory.
- Always return a short spoken 'say' field optimized for voice (< ~40 words) plus structured data for screens.
- Honesty Score changes must be explainable (ledger events with reason + delta).
- The product should feel slightly uncomfortable in a productive way, never hostile or condescending.
- Dismissing a challenge without engagement should have a visible (but not punitive) cost to the score.

5. Domain Knowledge the Developer Needs

Intellectual Honesty Concepts

Intellectual honesty is the willingness to update beliefs in response to evidence and argument, even when it is costly to one's identity or prior statements. Key related ideas a developer should understand: steelmanning vs strawmanning, motivated reasoning, confirmation bias, the illusion of explanatory depth, and the difference between feeling certain and being calibrated.

Why Steelmanning Matters

A weak or sarcastic counter-argument is easy to dismiss and teaches nothing. A high-quality steelman forces the user to confront the best version of the other side. The quality of the steelman is therefore a product feature, not an implementation detail. Bedrock prompts must be carefully engineered for this.

Scoring Philosophy

The Intellectual Honesty Score is not a moral grade. It is a private, longitudinal signal of engagement quality. Start at a neutral baseline (e.g. 50). Reward genuine engagement, quality of reply, completion of challenges, and written reflection. Do not heavily punish disagreement itself — only refusal to engage or low-effort dismissal. Keep the formula simple and transparent so the user can understand why the number moved.

Voice UX Constraints

- Spoken responses must be short and clear.
- Long steelmans should be summarized for speech and expanded on screen.
- Multi-turn debate needs explicit session state so the conversation can resume.
- Proactive challenges (challenge_me) should feel invitational, not nagging.

6. Data & State a Developer Must Care About

Concept	Why it matters	Typical fields
User / Device profile	Per-user honesty score and preferences	userId, score, preferences, topicHistory
Stance analysis	Input to steelman quality and scoring	statement, confidence, emotions, blindSpots
Challenge	Unit of intellectual work	id, topic, statement, intensity, status, reflection
Debate session	Multi-turn state across MCP calls	sessionId, turns[], qualityScores, maxTurns
Honesty event	Auditable score changes	eventId, delta, reason, relatedId, timestamp
Engagement record	Did the user actually grapple with the steelman?	engaged, quality, reply length/quality signals

Everything that changes the Honesty Score should be an append-only event. This makes the product explainable and debuggable.

7. Success Criteria & Anti-Goals

Product is succeeding when…

- A user can state a strong opinion and immediately receive a high-quality opposing case.
- The user can see their engagement history and score trend.
- The MCP tools are real, discoverable, and correctly implemented (not simulated in the frontend).
- Judges can run the demo in under two minutes and understand the thesis.
- The experience feels useful rather than preachy.

Anti-Goals (Do Not Build)

- A general chatbot that sometimes argues.
- A political fact-checker or ideology enforcer.
- A system that punishes disagreement itself.
- A complex multi-agent research system that is slow and opaque.
- A product that requires Alexa+ partner access to demo.

This document is the product source of truth. Every technical decision should be traceable back to the problem, the user value, and the non-negotiable rules above.

Echo Chamber Breaker — Refuse the comfortable opinion.